

Differential equations

Exercises Module 3

Exercise 1

Find all the partial derivatives of the following functions:

1. $f(x, y) = x^2 e^{2y}$
2. $f(x, y) = \ln(x + y^2)$
3. $f(x, y, z) = e^{xyz} + z$

Exercise 2

Given $z = xy^2$, with $x = s \cdot \cos(t)$ and $y = st$, calculate $\frac{\delta z}{\delta s}$ and $\frac{\delta z}{\delta t}$.

Exercise 3

Show that $y_1 = e^{-2x}$ and $y_2 = xe^{-2x}$ are both solutions of $y'' + 4y' + 4y = 0$.
is $y_3 = (x + 5)e^{-2x}$ also a solution?

Exercise 4

Solve the following differential equations:

1. $\frac{dy}{dx} = yx^2, y(0) = e$;
2. $\frac{dy}{dx} = \frac{y}{2x}, y(1) = 2$;
3. $\frac{dy}{dx} = x + 1, y(2) = 0$;
4. $\frac{dy}{dx} = xe^{-y}, y(2) = \ln(2)$
5. $\frac{dy}{dx} = 2 + e^x, y(0) = 5$
6. $\frac{dy}{dx} - e^y \cos(x) = 0, y(0) = 0$